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To leverage my 5 years of experience as a Data Engineer to contribute effectively to an organization where I can apply my expertise in designing, developing, and maintaining scalable data infrastructure and solutions.



SUMMARY

- 5 years of expertise designing, developing, and executing data pipelines and data lake requirements in numerous companies using the **Big Data Technology stack, Python, PL/SQL, SQL, REST APIs**, and the **Azure cloud platform**.
- Used **Spark Data Frame** operations to complete crucial data validations and **Spark Data Frame API** extensively on the **Cloudera infrastructure** to do analytics on **Hive** data.
- Hands-on experience developing data pipelines using **Spark components, Spark SQL, Spark Streaming** and **MLlib**. Hands-on experience with **Azure SQL Database, Azure SQL Data Warehouse, Azure Analysis Services, HDInsight, Azure Data Lake** and **Data Factory**.
- Designed, build and managed **ELT** data pipelines leveraging **Airflow, Python, and Google Cloud Platform** solutions. Hands-on experience interacting with **REST APIs** developed using the **micro-services** architecture for retrieving **data** from different sources.
- Has experience with story grooming, **Sprint** planning, daily stand-ups, and software techniques like **Agile** and **SAFe**. Experienced with **Dimensional modelling, Data migration, Data cleansing, Data profiling**, and **ETL Processes** features for **data warehouses**.
- Excellent problem solver possessing positive outlook with a blend of technical and managerial skills aimed to deliver the task before deadlines. Expert in creating various **Kafka** producers and consumers for seamless **data** streaming with **AWS** services.
- Practical knowledge in setting up and designing large-scale **data lakes, pipelines**, and effective **ETL (extract/load/transform)** procedures to collect, organize data that can be used to Converted a current application to use **Azure cloud** databases and storage.
- Experience with **Snowflake** cloud data warehouse and **AWS S3** bucket for integrating data from multiple source system which include loading nested **JSON** formatted data into **snowflake** table. Develop batch processing solutions by using **Data Factory** and **Azure Data bricks**.
- Experience on **Migrating SQL** database to **Azure data Lake, Azure data lake Analytics, Data Bricks** and **Azure SQL Data warehouse** and controlling and granting database access and Migrating On premise databases to **Azure Data Lake** store using **Azure Data factory**.
- Experience in **Windows Azure Services** like **PaaS, IaaS** and worked on storages like **Blob (Page and Block), SQL Azure**. Well experienced in deployment & configuration management and Virtualization. Deployed **Dockers** Engines in Virtualized Platforms for containerization of multiple apps.
- Expertise in building **CI/CD** on **AWS** environment using **AWS Code Commit, Code Build, Code Deploy** and Code Pipeline and experience in using **AWS CloudFormation, API Gateway**, and **AWS Lambda** in automation and securing the infrastructure on **AWS**.
- Experience in installation, configuration, supporting and managing - **Cloudera's Hadoop** platform along with **CDH3&4 clusters**.
- Highly skilled in using visualization tools like **Tableau, matplotlib, ggplot2** for creating dashboards. Strong **Data Analysis** skills using **Business intelligence, SQL** and **MS Office Tools**.
- Deep understanding of **SaaS business models** and **enterprise software product development**, contributing to efficient and scalable data solutions.
- Designed and optimized both **relational** and **non-relational databases** to enhance data storage and retrieval efficiency.
- Implemented **data warehousing** solutions like **Redshift** and **Snowflake** to manage large-scale data operations.
- Extensive hands-on experience with **Teradata** and its utilities, including **BTEQ, TPT, FastLoad, SQL queries**, and scheduling tools such as **Maestro** and **Control-M**.
- Familiarity with **Azure networking, security protocols**, and **key vault management**.
- Practical experience with **Redshift MPP databases**, focusing on performance optimization and tuning.

- Familiarity with **ITIL** processes, including Incident, Problem, and Change Management.
- Good understanding of **Linux Operating System** commands and hands-on experience in Oracle Database Administration.

PROFESSIONAL EXPERIENCE

Bank of America,
Azure Data Engineer

Columbia, Maryland, USA
June 2024–Present

- Developed methods for **Create, Read, Update and Delete (CRUD)** in Active Record. Worked on **Jenkins** pipelines to run various steps including **unit, integration** and **static analysis tools**.
- Worked on **SQL** and **PL/SQL** for backend data transactions and validations
- Developing **data pipelines** and workflows using **Azure Databricks** to process and transform large volumes of data, utilizing programming languages such as **Python, Scala, or SQL**. Written queries in **MySQL** and **Native SQL**.
- Build and deployed the code **Artefacts** into the respective environments in the Confidential **Azure cloud**. Experience in creating **Kubernetes** replication controllers, **Clusters** and label services to deployed **Microservices** in **Docker**.
- Used **Python** to write **Data** into **JSON** files for testing **Django** Websites, Created scripts for **data modelling** and **data import and export**.
- Used **Kafka** functionalities like **distribution, partition, replicated** commit log service log service for messaging systems by maintaining feeds. Involved in loading data from rest endpoints to **Kafka**.
- Managed **relational database** services in which the **Azure SQL** handles reliability, scaling, and maintenance. Integrated data storage solutions. Handled importing of data from various data sources, performed transformations using B, loaded data into **HDFS** and extracted the data from **SQL** into **HDFS** using **Sqoop**.
- Designed **GIT** branching strategies, merging per the needs of release frequency by implementing **GIT flow workflow** on **Bit bucket**. Led requirement gathering, business analysis, and technical design for **Hadoop** and **Big Data** projects.
- Integrated **Azure Data Factory** with **Blob Storage** to move data through **DataBricks** for processing and then to **Azure Data Lake Storage** and **Azure SQL** data warehouse. Working on **migrating Data** to the cloud (**Snowflake** and **AWS**) from the legacy data warehouses and developing the **infrastructure**.
- Enhanced by adding **Python XML SOAP** request/response handlers to add accounts, modify trades and security updates.
- Created **Data** stage and **ETL** jobs for populating the **data** into **Data Warehouse** constantly from different source systems like **ODS, flat files, Parquet**. Expertise in **Business intelligence** and **Data Visualization** tools like **Tableau**. Used **Tableau** to connect to various sources and build graphs.
- Converted and parsed data formats using **PySpark** Data Frames, reducing time spent on **data conversion** and parsing by 40%. Used **Azure Data Factory** to ingest data from log files and business custom applications processed data on Data bricks per day- to-day requirements and loaded them to **Azure Data Lakes**.
- Utilized **Elasticsearch** and **Kibana** for indexing and visualizing the real-time analytics results, enabling stakeholders to gain actionable insights quickly. Used **Python, R, SQL** to create **Statistical algorithms** involving **Multivariate Regression, Linear Regression, Logistic**. Worked on **Kafka** publishing the messages for further downstream systems.
- Implemented **data governance practices** to ensure data quality, security, and compliance across the data pipeline.
- Collaborated with cross-functional teams to define and gather data requirements, enhancing communication and project delivery.

Environment: Azure Data Factory, Apache Airflow, Talend, Spark, Azure SQL Database, Snowflake, SQL Server, Hive, T-SQL, Spark, Kafka, Azure Data Lake, Azure Synapse Analytics, Python, Scala, Tableau, Git, Jupyter Notebooks.

Boeing,
AWS Data Engineer

Annapolis Junction, Maryland, TX
July 2023–May 2024

- Implemented End to End solution for hosting the **web application** on **AWS cloud** with integration to **s3 buckets**.
- Developed a fully automated continuous integration system using **Git**, **Jenkins**, **MySQL** and custom tools developed in **Python** and **Bash** which saved \$85K YOY.
- Worked on **AWS Elastic Beanstalk** for fast deploying of various applications developed with **Java**, **PHP**, **NodeJS**, **Python** on familiar servers such as Apache.
- Developed server-side software modules and client-side user interface components and deployed entirely in Compute Cloud of **Amazon Web Services (AWS)**. Implemented **Lambda** to configure **Dynamo DB** Autoscaling feature and implemented **Data Access Layer** to access **AWS DynamoDB data**.
- Automated nightly build to run quality control using **Python** with **BOTOS** library to make sure pipeline does not fail which reduces the effort by 70%. Involved with development of Ansible playbooks with **Python** and **SSH** as wrapper for management of **AWS** node configurations and testing playbooks on **AWS instances**.
- Worked on **AWS Services** like **AWS SNS** to send out automated emails and messages using **BOTO3** after the nightly run.
- Worked on the development of tools which automate **AWS server** provisioning, automated application deployments, and implementation of basic failover among regions through **AWS SDK's**.
- Worked on **AWS CLI Auto Scaling** and **Cloud Watch Monitoring** creation and update.
- Allotted permissions, policies and roles to users and groups using **AWS Identity and Access Management (IAM)**.
- Created **AWS Lambda**, **EC2 instances** provisioning on **AWS** environment and implemented security groups, administered **Amazon VPC's**. Used **Jenkins** pipelines to drive all **micro-services** builds out to the **Docker** registry and then deployed to **Kubernetes**, **Created Pods** and managed using **Kubernetes**

Environment: Spark, Python, AWS, S3, Glue, Redshift, DynamoDB, Hive, SparkSQL, Docker, Kubernetes, Airflow, ETL workflows

Ivy Software Development Services
Private Limited (Ladbroke's)
 Data Engineer

Hyderabad, India
August 2020 – July 2022

- Analyzing the functional requirement documents from Business. Working knowledge on **Kubernetes** to deploy scale, **load balance**, and manage **Docker** containers and **Open Shift** with multiple namespace versions.
- Wrote and executed various **MySQL** database queries from **Python** using **Python-MySQL** connector and **MySQL dB** package. Developed data connectors to extract data from social media **APIs** and integrated them into the pipeline using **Python** and **Apache Kafka** Connect.
- Implemented airflow for workflow automation and scheduling tasks and created **DAGs** tasks.
- Responsible for Building and Testing of applications. Experience in handling **database** issues and connections with **SQL** and **NoSQL** databases like **MongoDB** by installing and configuring various packages in **Python** (**Teradata**, **MySQL**, **MySQL connector**, **PyMongo** and **SQLAlchemy**).
- Using **Azure Cluster** services, **Azure Data Factory V2** ingested a large amount and diversity of **data** from diverse source systems into **Azure Data Lake Gen2**. Used Continuous Delivery Pipeline.
- Deployed **microservices**, including provisioning **Azure** environments and developed modules using **Python Scripting** and **Shell Scripting**
- Imported real time weblogs using **Kafka** as a messaging system and ingested the **data** to **Spark Streaming** and did data quality checks using **Spark Streaming** and arranged bad and passable flags on the **data**.
- Involved in various phases of **Software Development Lifecycle (SDLC)** of the application, like gathering requirements, design, development, deployment, and analysis of the application.
- Consult leadership/stakeholders to share design recommendations and thoughts to identify product and technical requirements, resolve technical problems and suggest **Big Data** based analytical solutions.
- Developed **Spark** applications using **Scala** and **Spark SQL** for **Data Extraction**, **transformation** and **aggregation** from multiple file formats for analysing and transformation the **data** uncover insight into the customer usage patterns and even.

- Build **Jenkins** jobs for **CI/CD** infrastructure for **GitHub repos**. Involved in loading and transforming large sets of structured, Semi-structured end unstructured data and analyzed them by running **Hive** queries. Processed the image data through the **Hadoop** distributed system by using **Map** and **Reduce** then stored into **HDFS**.
- Designing and implementing data integration solutions using **Azure Data Factory** to move data between various data sources, including on-premises and cloud-based systems.
- Controlling and granting database access and migrating on premise databases to **Azure data lake** store using **Azure Data Factory**. Developed remote integration with third-party platforms by using **RESTful** web services.

Environment: Python, Django, JavaScript, MySQL, NumPy, SciPy, PandasAPI, PEP, PIP, Jenkins, JSON, Git, JavaScript, AJAX, RESTful web service, MySQL, PyUnit.

RAKUTEN,
Data Analyst

Bangalore, India
March 2018 – July 2020

- Executed advanced analytics functions, including proficient data modelling and prescriptive analytics.
- Applied techniques and business intelligence tools such as **Tableau** and **PowerBI** to craft interactive dashboards and ad hoc reports, effectively addressing business challenges and streamlining operations.
- Created **ad hoc** queries using **SQL** within **Cognos Report Net**, meticulously examining reports, and conveying findings through impactful presentations in **Power Point** and **Excel**.
- Revitalized standard operational procedures, driving improvements in functional areas and data management practices.
- Maintained a remarkable 97% on-time delivery rate for reports, ensuring timely and accurate insights.
- Facilitated **seamless** data distribution, streamlined file sharing, and maintained meticulous **report tracking**.
- Utilized **Python** to create clusters for identifying in active clients, offering valuable insights to marketing and sales teams.
- Worked with Business users during **requirements gathering** and business analysis prepare high level **Logical Data Models** and **Physical Data Models** and Performed Reverse Engineering of the current application
- Analysis of **functional** and **non-functional** categorized data elements for data profiling and mapping from source to target data environment and Written **SQL** scripts to test the mappings and Developed **Traceability Matrix** of Business Requirements mapped to **Test Scripts** to ensure any **Change Control** in requirements leads to test case update
- Involved in integration of various relational and non-relational sources such as **DB2, Teradata, Oracle, SFDC, Netezza, SQL Server, COBOL, XML and Flat Files**. Redefined many attributes and relationships in the reverse engineered model and cleansed unwanted tables columns on **Teradata** database as part of data analysis responsibilities
- Expertise in using data modelling tools like **MS Visio** and **ER/Studio Tool** for logical and physical design of databases
- Performed Data Modelling, Database Design, and Data Analysis with the extensive use of **ER/Studio** and documented **ER Diagrams, Logical** and **Physical** models, business process diagrams and process flow diagrams
- Created reports in **Oracle Discoverer** by importing **PL/SQL** functions on the Admin Layer, in order to meet the sophisticated client requests

Environment: Python, Pandas, NumPy, Tableau, Power BI, SQL, MS Excel, Data Distribution, File Sharing, Identifying Data Issues, Reducing Redundancy.

EDUCATION

UNIVERSITY OF MARYLAND, USA
Master's in Data Science

TECHNICAL SKILLS

Programming Languages	Python (NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn), R (ggplot2, dplyr), SQL, MATLAB, Java, PowerShell, Bash, Korn Shell, AWK
Machine Learning & AI	Supervised & Unsupervised Learning, Deep Learning (CNN, RNN, LSTM, Transformers), NLP (BERT, GPT, NLTK, SpaCy, Gensim), Time Series Forecasting, Model Optimization, ML Pipelines, Recommendation Systems
Big Data & Cloud	Apache Spark, Snowflake, Hadoop, Kafka, Flink, Google BigQuery, AWS (EC2, S3, Redshift, SageMaker, Lambda, Glue), Microsoft Azure (Synapse, Databricks, Data Factory), GCP (BigQuery, AI Platform)
Statistical & Analytical Methods	Hypothesis Testing, ANOVA, PCA, Bayesian Inference, A/B Testing, Correlation Analysis, Feature Engineering, Dimensionality Reduction
Data Engineering & ETL	Data Pipelines, Apache Airflow, Spark Streaming, Hadoop Ecosystem (HDFS, MapReduce, Hive, Sqoop, Pig), ETL Processing (SSIS, Talend)
Data Warehousing & Databases	PostgreSQL, MS SQL Server, MongoDB, MySQL, Oracle, Teradata, Redshift, Snowflake, HBase
Data Visualization & BI	Tableau, Power BI, QlikView, MicroStrategy, Python (Matplotlib, Seaborn, Plotly), R (ggplot2, Shiny), D3.js
Model Deployment & DevOps	FastAPI, Flask, Kubernetes, Docker, CI/CD (Jenkins, GitHub Actions), Serverless Computing